THIRUKUMARAN T

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY:

M.Pharm in Pharmaceutical Biotechnology (CGPA 9.42/10, Tata Trusts Scholar) with expertise in genomic data analysis and computational biology. Proficient in scRNA-seq analysis, differential expression studies, and bioinformatics pipeline development (R, Seurat, Docker). Passionate about leveraging computational approaches to advance drug discovery and precision medicine.

PROJECTS:

Tumour microenvironment characterisation in glioblastoma (scRNA-seq analysis):

- Characterised the full cellular landscape of human glioblastoma using single-cell transcriptomics: resolved 12 cell populations (2,847 cells; 24,381 genes; 10x Genomics) including clinically relevant malignant subtypes -OPC-like, mesenchymal-like, and hypoxia-associated - each with distinct therapeutic implications.
- Developed a reproducible pipeline (R, Seurat, Docker, GitHub Actions) translatable to clinical NGS datasets for tumour subtype stratification.
- Published detailed research blog: [https://fromreadstoresultsttk.blogspot.com/] | **GitHub** - https://github.com/TTK-Pharma/NGS-data-analysis
- **Tools:** R (Seurat, DESeq2, Bioconductor), GitHub

Biopharmaceutical Characterization and Functional Potency Assessment of Poliovirus-Like Particles (PVLP)

- Executed analytical characterization of 5 vaccine batch samples using HCP, HCD, and SDS-PAGE assays to evaluate protein quality and safety per regulatory standards.
- Analyzed and interpreted experimental data across multiple assays; identified quality parameters within WHO and regulatory compliance guidelines.
- Contributed to assay validation protocols and generated technical documentation for regulatory submissions.
- Published more informative research blog: [https://thirukumaranrtm.blogspot.com/]
- **Tools & Techniques:** ELISA, PicoGreen Assay, BCA Assay, SDS-PAGE, SNT, Cell Culture Techniques

CLI tool for reproducible scRNA-seq analysis | Tools: R, Seurat, Docker | [In Development]

- Developed a configurable CLI pipeline for scRNA-seq analysis (R, Seurat, Docker) designed for reproducibility across 10x Genomics datasets — applicable to clinical tumour profiling workflows.
- Implements QC, normalisation, clustering, and visualisation via user-defined parameters.

SKILLS:

Bioinformatics & Genomics:

- scRNA-seq analysis
- Differential Expression Analysis
- Variant Analysis
- Sequence Analysis

Programming & Data Analysis:

- R
- Python
- Docker, Git, Linux
- SQL
- MS Office

Bioanalytical assay:

- HCP (Host Cell Protein) assay
- HCD (Host Cell DNA) assay

Cell Culture:

- Cell line maintenance
- Cell Passaging

- BCA assay
- D-AG (D-Antigen) assay
- SNT (Serum Neutralization Test)
- Sub culturing
- MTT assay
- Wound Healing assay

Molecular biology:

- SDS PAGE
- Western Blotting
- PCR
- cDNA synthesis

PROFESSIONAL EXPERIENCE:

Biological E Limited | Hyderabad Oct 2025 – Mar 2026

Trainee | R&D-Analytical Development

- Performed analytical characterization of **5 vaccine batch samples** using **HCP, HCD, and SDS-PAGE assays** to evaluate protein quality and safety parameters
- Generated and interpreted experimental data for **protein quality and safety assessment**
- Contributed to **assay validation and regulatory compliance documentation**

Fourrts India Laboratories Pvt. Ltd| Chennai | Intern Jan 2023 – Feb 2023

- Gained exposure to pharmaceutical manufacturing and industrial operations.
- Observed quality control, production, and formulation processes in a regulated pharmaceutical environment.
- Learned about GMP practices, documentation procedures, and workflow management in the pharmaceutical industry.
- Developed understanding of industrial pharmaceutical processes and regulatory compliance.

ACADEMIC CREDENTIALS:

Master of Pharmacy (Pharmaceutical biotechnology) 2024 – 2026

JSS College of Pharmacy, Ooty | CGPA-9.42/10 | Tata Trusts Scholarship

Bachelor of Pharmacy 2018 – 2023

Arulmigu Kalasalingam College of Pharmacy, Krishnankovil | CGPA-7.4/10

COURSES AND CERTIFICATES:

- E-Poster presentation on "Molecular Characterization of APOE gene SNPs in Alzheimer's Disease using Comparative genomics and population-based analysis" and organizing committee member for international Conference "PHARMANCIA" | JSS College of Pharmacy, Ooty | 7 & 8 March 2025.
- Successfully completed the NPTEL online course and exam on intellectual property.
- Workshop on stem cell research | EDUTECH LIFE-MECHANICA IIT Madras | February 22-23, Chennai.
- Internship on Basic Cell Culture Techniques | International Research Centre, Sathyabama University | January 03-13 2025.
- Completed the training program of **Full Stack Web Development (MERN)** from Edureka

PUBLICATIONS:

Article titled "*Precision Probiotics for Healthy Aging: Toward Personalized Gerobiotic Therapeutics*" *Under review (Manuscript submitted)*.

LANGUAGES:

- English (Full working proficiency), Tamil (Native speaker), Telugu (Working proficiency)